Methodological Report on Data Auditing, Cleaning, and Imputation – Retail Store Sales
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Fuente: https://www.kaggle.com/code/mahmoudredagamail/retail-store-sales-dirty-for-data-cleaning

1. Introduction

The project uses the Retail Store Sales: Dirty for Data Cleaning dataset, which consists of 12,575 transactions and 11 variables related to customers, products, categories, prices, quantities, total spending, payment methods, locations, transaction dates, and discount application. The dataset was designed to contain data quality issues, providing an opportunity to develop a systematic process for data auditing, cleaning, and imputation.

Data processing was divided into two stages. First, a data quality audit and deterministic cleaning based on business rules were performed using Python. Subsequently, R was used to assess missing values that could not be recovered exactly and determine which could be addressed through statistical imputation methods.

The methodological approach focused on preserving the original data and modifying values only when a validated deterministic rule or sufficient statistical evidence supported the imputation.

2. Initial Dataset Structure

The Retail Store Sales: Dirty for Data Cleaning dataset consists of 12,575 records and 11 variables containing information about retail transactions, including customer and transaction identifiers, product categories, purchased items, unit prices, quantities sold, total spending, payment methods, locations, purchase dates, and discount application.

The variables were classified according to their function into two identifiers (Transaction ID and Customer ID), four categorical variables (Category, Item, Payment Method, and Location), three numerical variables (Price Per Unit, Quantity, and Total Spent), one temporal variable (Transaction Date), and one binary variable (Discount Applied). This classification helped identify the characteristics of the data and establish appropriate validation criteria for each variable type.

The three numerical variables describe the economic components of the transactions and allow an assessment of the consistency between unit price, quantity purchased, and total spending. This relationship facilitates the identification of inconsistencies and the recovery of certain missing values through business rules, provided that the relationships between variables are verified and the potential effect of discounts on the recorded transaction amount is considered.

The Transaction Date variable was initially stored as text and was subsequently converted to a date format to facilitate temporal analysis. The Discount Applied variable indicates whether a discount was applied and requires a consistency check of its encoding, as it initially contained Boolean or text values.

The review of this structure helped establish the initial criteria for the data quality audit, identify relationships that could be used in deterministic data cleaning, and distinguish values recoverable through verifiable rules from those requiring statistical assessment before imputation could be considered.

Variable	Initial type	Description
Transaction ID	Text	Transaction identifier
Customer ID	Text	Customer identifier
Category	Text	Product category
Item	Text	Purchased product
Price Per Unit	Numeric	Unit price
Quantity	Numeric	Quantity purchased
Total Spent	Numeric	Total amount spent
Payment Method	Text	Payment method used
Location	Text	Purchase channel or location
Transaction Date	Text	Transaction date
Discount Applied	Boolean-Text	Indicates whether a discount was applied

Table 1. Dataset variables
Source. https://www.kaggle.com/code/mahmoudredagamail/retail-store-sales-dirty-for-data-cleaning

Table 1 presents the dataset structure, consisting of 12,575 observations and 11 variables classified according to their function as identifiers, categorical, numerical, temporal, and binary variables. The dataset primarily contains transactional and categorical information. Among the numerical variables, Price Per Unit, Quantity, and Total Spent allow the consistency of recorded purchase amounts to be assessed and relationships to be established for data validation. The Transaction Date variable was initially stored as text and subsequently converted to a date format to facilitate temporal analysis. Overall, this structure supports the identification of potential inconsistencies and the application of business rules for data cleaning and missing value recovery, provided that the underlying relationships have been verified.

3. Initial Missing Data Assessment

The initial data quality audit identified five variables with missing values, revealing differences in the completeness levels of the recorded information.

Variable	Missing Values	Percentage
Discount Applied	4.199	33,39 %
Item	1.213	9,65 %
Price Per Unit	609	4,84 %
Quantity	604	4,80 %
Total Spent	604	4,80 %

Table 2. *Missing Values Before Cleaning*
Source. *Prepared by the author.*

Table 2 presents the distribution of missing values across five variables in the original dataset. Discount Applied has the highest proportion of missing data, with 4,199 observations (33.39%), followed by Item, with 1,213 (9.65%). The numerical variables Price Per Unit, Quantity, and Total Spent have missing data

percentages close to 5%. The identical count of 604 missing values in Quantity and Total Spent suggests a possible shared missingness pattern, although further verification is required to determine whether these values occur in the same transactions. These findings identify the main data completeness issues and guide the subsequent analysis of missingness patterns before selecting appropriate recovery and imputation procedures.

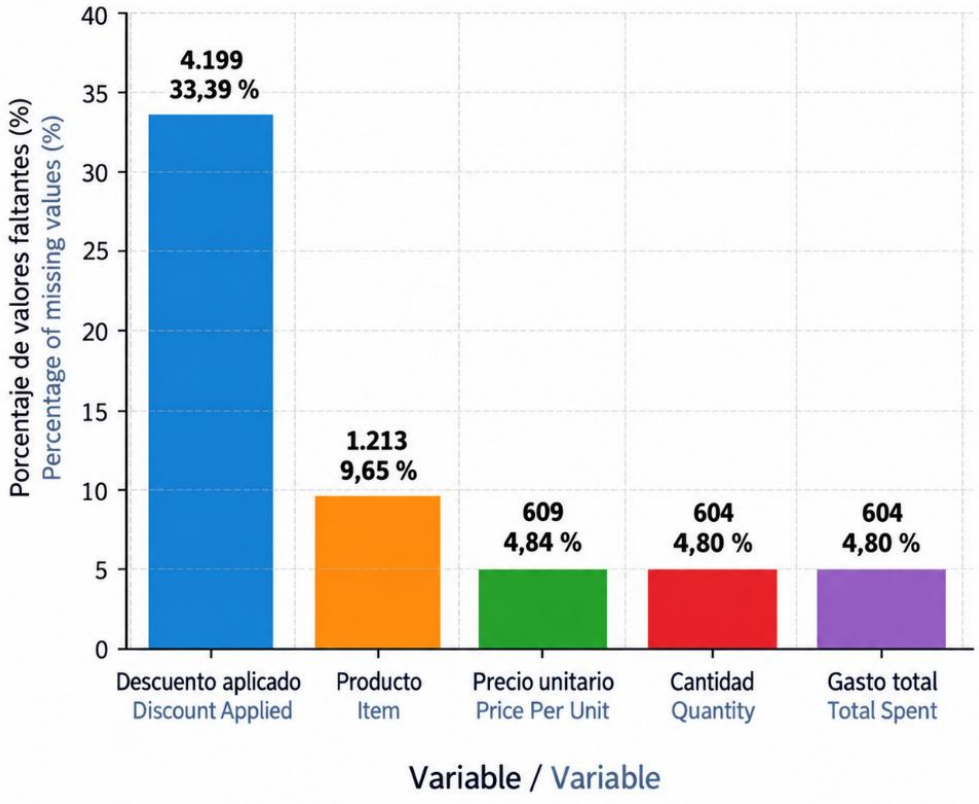


Figure 1. *Distribution of Missing Values by Variable in the Original Dataset.*
Source. *Prepared by the author.*

Figure 1 presents the number and percentage of missing values across five variables in the original dataset, which contains 12,575 transactions. The Discount Applied variable has the highest proportion of missing data, with 4,199 observations (33.39%), followed by Item, with 1,213 (9.65%). The numerical variables Price Per Unit, Quantity, and Total Spent show considerably lower percentages, ranging from 4.80% to 4.84%. In particular, Quantity and Total Spent each contain 604 missing values, suggesting a possible shared missingness pattern. However, this coincidence alone does not confirm that the missing values occur in the same transactions. These findings reveal differences in data completeness across variables and provide a basis for examining missingness patterns before applying deterministic recovery procedures or statistical imputation methods.

4. Cardinality of the variables

The number of distinct values contained in each variable was also studied.

Variable	Unique Values
Discount Applied	2
Location	2
Payment Method	3
Category	8
Quantity	10
Customer ID	25
Price Per Unit	25
Item	200
Total Spent	227
Transaction Date	1.114
Transaction ID	12.575

Table 3. *Number of Unique Values by Variable in the Original Dataset.*
Source. *Prepared by the author.*

Table 3 presents the number of unique values identified in each variable of the original dataset. Transaction ID contains 12,575 distinct values, matching the total number of transactions and indicating that there are no duplicate transaction identifiers. In contrast, Customer ID contains 25 distinct identifiers, while Item includes 200 products and Category comprises 8 categories. Both Discount Applied and Location have two unique values, whereas Payment Method contains three. Among the numerical variables, Quantity has 10 distinct values, Price Per Unit has 25, and Total Spent has 227. Finally, Transaction Date contains 1,114 distinct dates. These findings provide an overview of variable cardinality, help identify potential coding inconsistencies, and guide the validation procedures required during data cleaning. However, the number of unique values alone is insufficient to determine whether the recorded categories or values are valid.

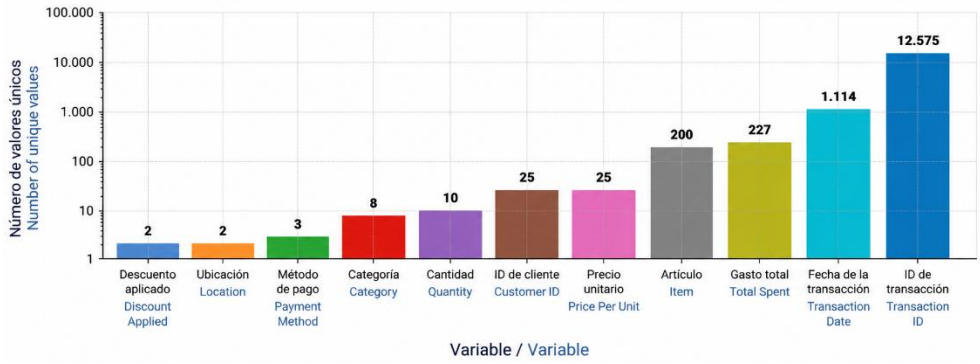


Figure 2. *Number of Unique Values by Variable in the Original Dataset.*
Source. *Prepared by the author.*

Figure 2 shows the number of unique values recorded for each variable, using a logarithmic scale to facilitate comparisons across variables with different

cardinality levels. Transaction ID contains 12,575 distinct values, matching the total number of records, while Transaction Date includes 1,114 unique dates. Among the purchase-related variables, Total Spent has 227 unique values, Item contains 200, and both Price Per Unit and Customer ID have 25 each. Meanwhile, Quantity contains 10 distinct values and Category has 8, whereas Payment Method, Location, and Discount Applied show lower diversity, with 3, 2, and 2 values, respectively. This distribution highlights differences in variable cardinality and helps guide the identification of potential inconsistencies during data cleaning. However, the number of unique values alone is insufficient to establish data validity or rule out duplicate records based on other combinations of variables.

5. Validation of the Quantity domain

Quantity represents the number of units purchased.

Quantity	Frequency
1	1.147
2	1.164
3	1.224
4	1.155
5	1.228
6	1.220
7	1.227
8	1.226
9	1.148
10	1.232
Missing item	604

Table 4. *Frequency Distribution of the Number of Units Purchased per Transaction in the Original Dataset.*
Source. *Prepared by the author.*

Table 4 presents the frequency distribution of the Quantity variable, which represents the number of units purchased per transaction. Observed values range from 1 to 10 units, with relatively similar frequencies. Transactions involving 10 units have the highest frequency, with 1,232 records, while transactions involving one unit have the lowest, with 1,147. This difference of 85 records indicates a relatively uniform distribution across the observed quantities, with no pronounced concentration at any particular value. Additionally, 604 missing observations were identified (4.80% of the dataset). Recovering these values requires assessing their relationships with other variables, particularly Price Per Unit and Total Spent, before considering statistical imputation procedures.

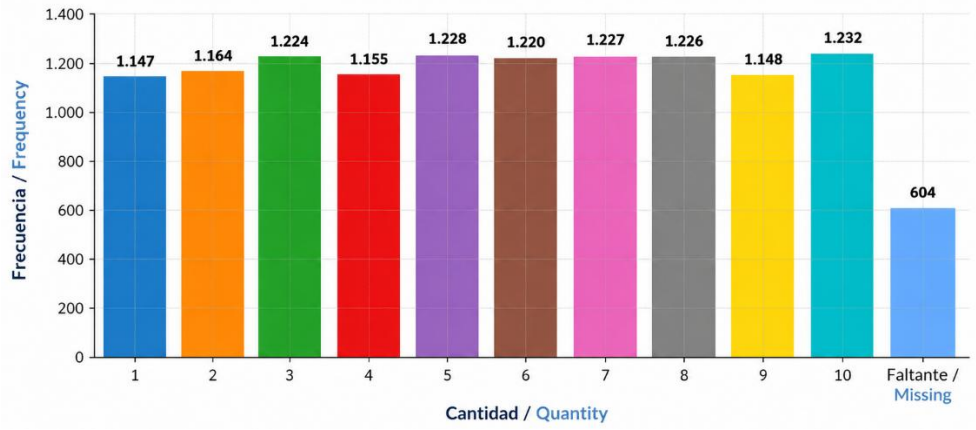


Figure 3. *Frequency Distribution of the Number of Units Purchased per Transaction in the Original Dataset.*
Source. *Prepared by the author.*

Figure 3 presents the frequency distribution of the Quantity variable, with observed values ranging from 1 to 10 units per transaction. The frequencies show a relatively uniform pattern, ranging from 1,147 to 1,232 records. Purchases of 10 units have the highest frequency, with 1,232 transactions, while purchases of one unit have the lowest, with 1,147. The difference of 85 observations indicates no pronounced concentration at any particular quantity. Additionally, 604 records contain missing values (4.80% of the dataset), limiting a complete analysis of purchased quantities. These findings provide a reference for subsequently evaluating the distribution of Quantity and determining whether recovery or imputation procedures alter its original characteristics.

6. Price Per Unit Validation
Exactly 25 different price levels were found, ranging from 5.0 to 41.0.

Indicator	Value
Minimum price	5,0
Maximum price	41,0
Number of distinct prices	25
Missing values	609
Values outside the expected range	0

Table 5. *Data Quality Indicators and Characteristics of the Price Per Unit Variable in the Original Dataset.*
Source. *Prepared by the author.*

Table 5 presents the results of the initial assessment of the Price Per Unit variable, which represents the unit price of purchased products. Observed values range from 5.0 to 41.0, with 25 distinct prices and no records outside the expected range. However, 609 missing values were identified, accounting for 4.84% of the 12,575 transactions. Although the recorded prices fall within the established validation limits, missing information affects the completeness of this variable. These findings help guide the recovery of unit prices through verifiable relationships with other variables, particularly Item, Quantity, and Total Spent, before considering statistical imputation methods.

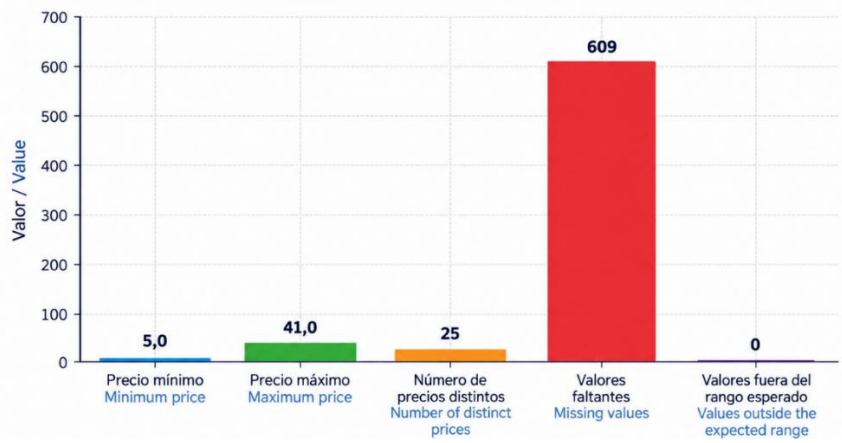


Figure 4. *Summary of Data Quality Indicators for the Price Per Unit Variable in the Original Dataset.*
Source. *Prepared by the author.*

Figure 4 presents the main indicators of the Price Per Unit variable, with observed values ranging from a minimum of 5.0 to a maximum of 41.0 and a total of 25 distinct unit prices. No values were identified outside the expected range, indicating that the available prices meet the established validation limits. However, 609 missing values were recorded, representing 4.84% of the transactions and affecting data completeness. These findings highlight the need to recover missing unit prices through verifiable relationships with other variables, particularly Item, Quantity, and Total Spent, before considering statistical imputation procedures.

7. Validation of business rules

Three relationships were verified before retrieving information.

Rule	Relationship
1	Item → Category
2	Item → Price Per Unit
3	Total Spent = Price Per Unit × Quantity

Table 6. *Business Rules Evaluated for Data Validation and Recovery.*
Source. *Prepared by the author.*

Table 6 presents the three business rules considered for checking record consistency and recovering missing values through deterministic procedures. The first establishes a relationship between the purchased item (Item) and its category (Category), while the second links each item to its unit price (Price Per Unit). Both relationships support data recovery when each item is verified to have a unique category and unit price among valid records. The third rule relates total spending (Total Spent) to unit price and quantity purchased (Quantity), allowing missing values to be calculated when the other two variables are available. Applying this rule requires verifying that total spending does not include additional adjustments, such as discounts. These rules provide a basis for recovering information before resorting to statistical imputation methods.

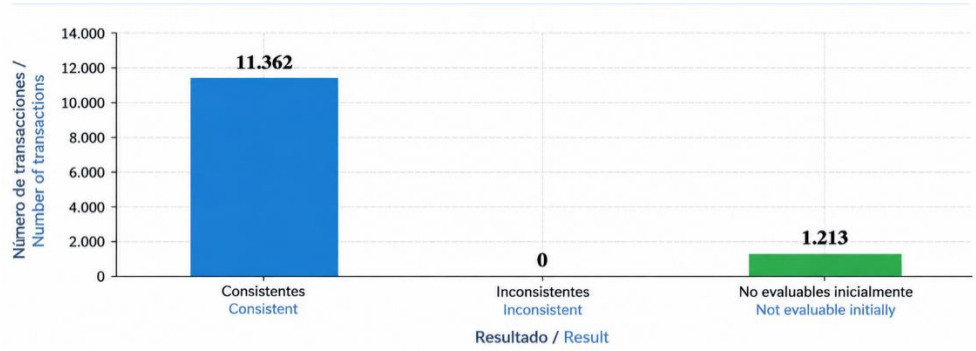


Figure 5. Validation of the Relationship Between Unit Price, Quantity, and Total Spending Using Business Rules.
Source. *Prepared by the author.*

Figure 5 presents the results of validating the relationship between Price Per Unit, Quantity, and Total Spent across the 12,575 transactions in the original dataset. Of these, 11,362 records (90.35%) could be evaluated and satisfied the established mathematical relationship, with no inconsistencies identified among the verifiable transactions. However, 1,213 records (9.65%) could not initially be evaluated because the information required to verify this relationship was missing. These findings support the application of business rules to recover certain missing values, although the observed consistency does not, by itself, guarantee data accuracy. The non-evaluable transactions require further processing to determine which values can be recovered through deterministic procedures and which require statistical imputation methods.

8. Missing Data Patterns

The missing data analysis identified four patterns of missing information in the original dataset.

Missing Data Pattern	Records
Complete records	7.579
Only Discount Applied missing	3.783
Item + Price Per Unit missing	609
Item + Quantity + Total Spent missing	604
Total	12.575

Table 7. *Distribution of Records by Missing Data Patterns in the Original Dataset.*
Source. *Prepared by the author.*

The analysis of missing data patterns identified four distinct situations across the 12,575 transactions in the original dataset. Of these, 7,579 records (60.27%) contain complete information, while 3,783 (30.08%) have missing values only in Discount Applied. Additionally, 609 transactions (4.84%) have simultaneous missing values in Item and Price Per Unit, and another 604 (4.80%) lack information in Item, Quantity, and Total Spent. This distribution shows that missing values occur in specific combinations of variables, allowing the identification of joint missingness patterns and guiding deterministic data recovery and statistical imputation procedures. However, these findings alone are insufficient to determine the statistical mechanism underlying the missing data.

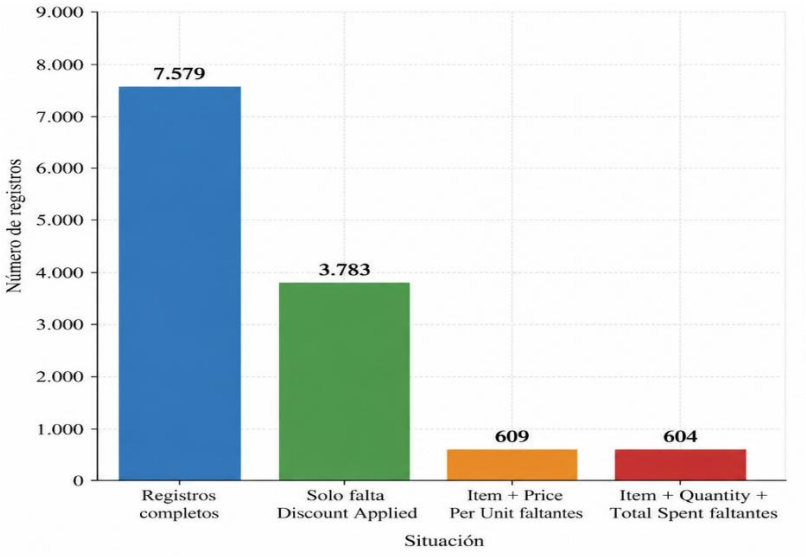


Figure 6. *Distribution of the Main Missing Data Patterns in the Original Dataset.*
Source. *Prepared by the author.*

Figure 6 shows the distribution of the main missing data patterns identified across the 12,575 transactions. Complete records account for the largest proportion, with 7,579 observations (60.27%), followed by 3,783 transactions (30.08%) with missing information only in Discount Applied. Additionally, 609 records (4.84%) have simultaneous missing values in Item and Price Per Unit, while another 604 (4.80%) contain missing values in Item, Quantity, and Total Spent. This distribution confirms the presence of joint missingness patterns in specific variables, particularly those related to products and the monetary values of transactions. Identifying these patterns helps guide the application of business rules for data recovery and determine which records require further statistical assessment.

9. Item Recovery

The first procedure used the relationship: Category + Price Per Unit → Item.
When this combination identified a unique product, the Item was retrieved.

Stage	Items Recovered
First recovery	604
Second recovery	609
Total recovered	1.213
Remaining missing values	0

Table 8. *Recovery of Missing Values in the Item Variable Through Deterministic Procedures.*
Source. *Prepared by the author.*

Table 8 presents the results of the missing value recovery process for the Item variable, conducted in two stages. In the first stage, 604 records were recovered, followed by an additional 609 in the second stage, resulting in a total of 1,213 recovered values. This procedure completed the Item information across all 12,575 transactions, reducing missing values from 1,213 (9.65%) to zero. The results indicate that all missing values were recovered using the applied rules, without requiring statistical imputation for this variable. However, achieving complete data

coverage does not, by itself, guarantee the accuracy of the recovered values, which depends on the validity of the business rules used.

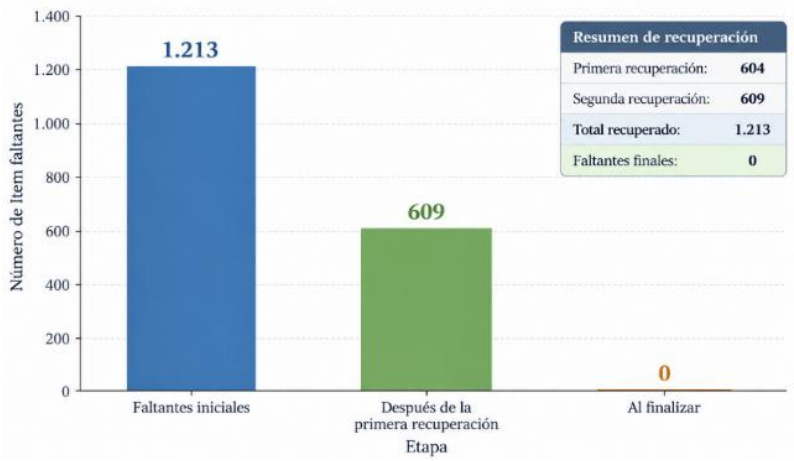


Figure 7. *Progressive Recovery of Missing Values in the Item Variable Through Deterministic Procedures.*
Source. *Prepared by the author.*

Figure 7 illustrates the progressive reduction of missing values in the Item variable, which initially contained 1,213 records with missing information (9.65% of the dataset). During the first stage, 604 values were recovered, reducing the number of missing entries to 609. Subsequently, following the recovery of Price Per Unit, the remaining 609 items were identified, completely eliminating missing values. This procedure achieved 100% completeness in Item without requiring statistical imputation methods. These findings illustrate the value of applying business rules sequentially, using information recovered in earlier stages to support subsequent recovery. However, the validity of the reconstructed values depends on the consistency and verification of the underlying relationships.

10. Price Per Unit Recovery

For the remaining 609 records without a price, Quantity and Total Spent were available. Therefore: Price Per Unit = Total Spent / Quantity.

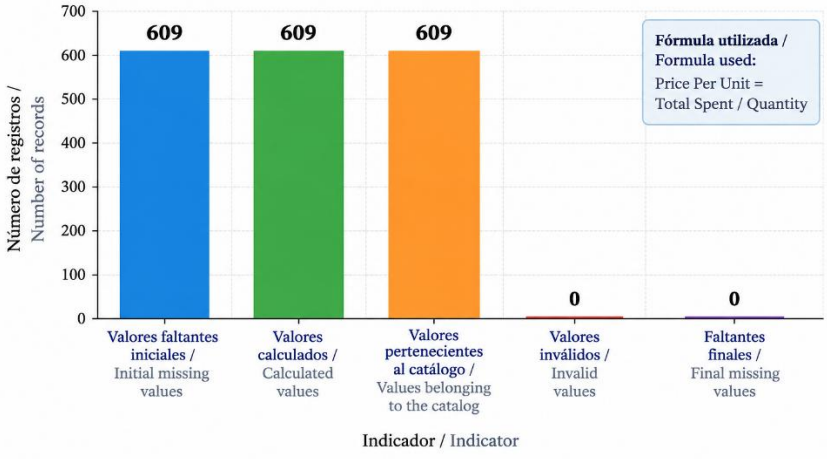


Figure 8. *Results of Missing Value Recovery in the Price Per Unit Variable Using Business Rules.*
Source. *Prepared by the author.*

Figure 8 presents the results of recovering 609 missing values in the Price Per Unit variable, representing 4.84% of the original transactions. Missing unit prices were

calculated using the relationship between total spending (Total Spent) and quantity purchased (Quantity), yielding 609 values that were subsequently checked against the valid price catalog. Validation confirmed that all recovered prices belonged to the set of permitted values, with no invalid results or remaining missing values after the procedure. Consequently, 100% completeness was achieved for this variable without statistical imputation. These findings support the application of business rules for data recovery when relationships between variables have been previously verified, although membership in the valid price catalog alone does not guarantee the accuracy of each reconstructed unit price.

11. Results of Deterministic Data Cleaning

Después de aplicar las reglas de negocio, se evaluó la reducción de los valores faltantes en cada variable. / After applying the business rules, the reduction in missing values was assessed for each variable.

Variable	Before	After	Recovered	Recovery Rate
Item	1.213	0	1.213	100 %
Price Per Unit	609	0	609	100 %
Quantity	604	604	0	0 %
Total Spent	604	604	0	0 %
Discount Applied	4.199	4.199	0	0 %

Table 9. *Comparison of Missing Values Before and After Deterministic Data Cleaning.*
Source. *Prepared by the author.*

Table 9 presents the results of deterministic data cleaning by comparing missing values before and after the procedure. All 1,213 missing values in Item and 609 in Price Per Unit were recovered, achieving a 100% recovery rate for both variables. In contrast, Quantity and Total Spent retained their 604 missing values each, while Discount Applied remained incomplete in 4,199 records. These findings indicate that business rules successfully recovered all missing product and unit price information but were insufficient to resolve missing values in the remaining variables. Therefore, the unresolved values require further statistical assessment to determine whether imputation is appropriate, considering the missingness patterns and available information.

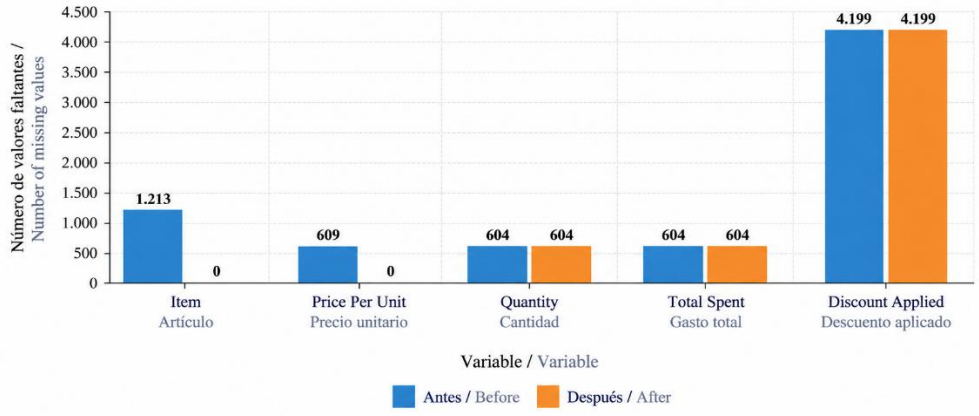


Figure 9. *Comparison of Missing Values Before and After Deterministic Cleaning.*
Source. *Prepared by the author.*

Figure 9 shows two bars for each variable: one representing the number of missing values before cleaning and the other representing the number of missing values after the process. This comparison shows that the reduction was complete for Item and Price

Per Unit. In Item, missing values decreased from 1,213 to 0, while in Price Per Unit, they decreased from 609 to 0. In contrast, Quantity remained at 604 missing values, Total Spent also remained at 604, and Discount Applied remained at 4,199.
These results do not indicate a failure of the cleaning process. Instead, they show that the business rules applied were able to recover only those values that could be reconstructed exactly and verifiably. In total, 1,822 deterministic transformations were performed, corresponding to the recovery of 1,213 values in Item and 609 in Price Per Unit.

12. Situation prior to statistical imputation

After the cleaning performed in Python, only three variables with missing data remained.

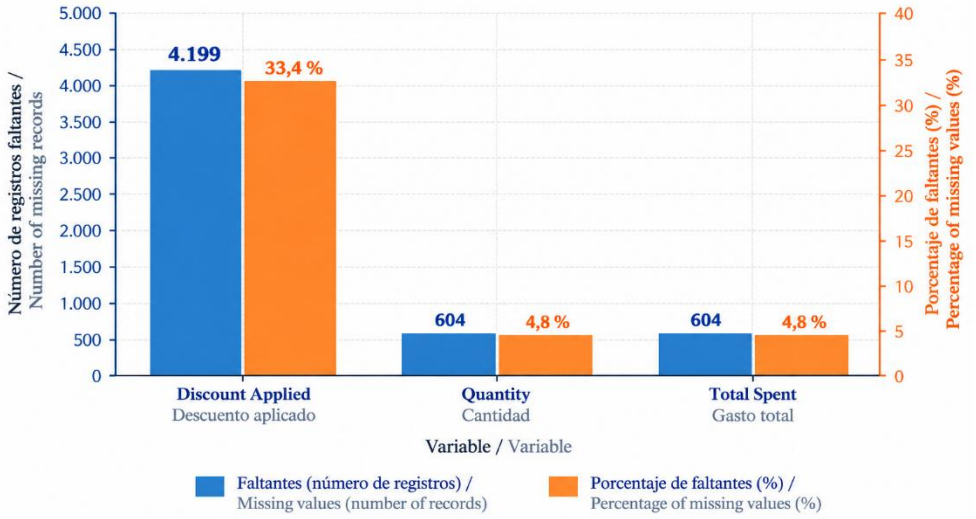


Figure 10. *Distribution of Missing Values Before Statistical Imputation.*
Source. *Prepared by the author.*

Figure 10 presents the missing values remaining after deterministic data cleaning and before applying statistical imputation procedures. The Discount Applied variable accounts for the largest amount of missing information, with 4,199 records (33.4%), while Quantity and Total Spent each contain 604 missing values (4.8%). Together, these three variables account for 5,407 missing values. The persistence of these missing values indicates that the applied business rules could not recover the information exactly, making it necessary to assess alternative statistical treatment methods. In particular, imputing Quantity requires considering its distribution and relationships with other variables, while handling Total Spent must preserve consistency between unit price, quantity, and total spending. For Discount Applied, the available information must be evaluated before determining whether statistical imputation is justified.

13. Preparation for Quantity Imputation

To impute missing values in the Quantity variable, five predictors were selected: Item, Payment Method, Location, year, and month. These variables provide information about the purchased product, payment method, purchase channel, and temporal characteristics of the transactions. The completeness assessment confirmed that all five predictors were fully observed, with no missing values. This allowed them to be included in the imputation procedure without requiring additional missing data treatment. However, their completeness does not guarantee predictive performance, making it necessary to assess their relationships with Quantity and the suitability of the selected statistical method.

14. Construction of the Validation Sample

To evaluate imputation methods for the Quantity variable, a validation sample was constructed by artificially masking originally observed values. Of the 11,971 records with available information, 1,197 observations, representing approximately 10%, were selected and temporarily masked to assess each method's ability to recover known values. These cases were combined with the 604 originally missing values, resulting in a total of 1,801 records with missing information during the validation procedure. This approach allowed imputed quantities to be compared with the actual values of the masked observations, providing a basis for measuring imputation error and selecting the best-performing method according to the established evaluation criteria. However, the applicability of these results to genuinely missing observations depends on how representative the validation sample is of records with missing information.

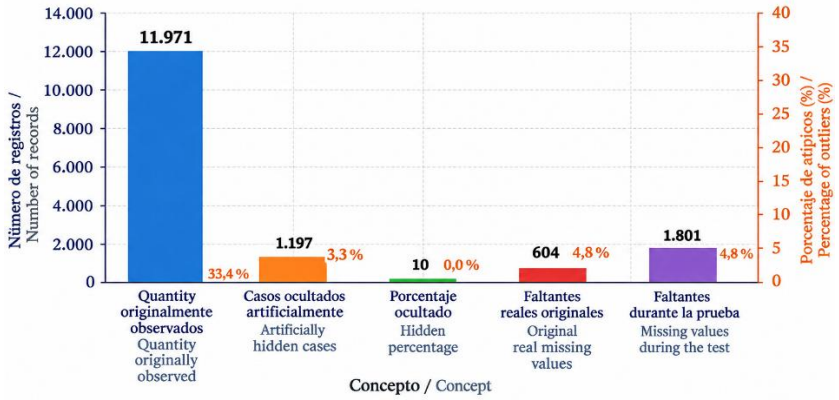


Figure 11. *Validation Design for the Imputation of the Quantity Variable.*
Source. *Prepared by the author.*

Figure 11 shows the design used to validate the imputation methods applied to the Quantity variable. Since it is not possible to calculate the error directly on the 604 truly missing records, because their actual values are unknown, a validation strategy based on the temporary masking of observed data was used. In total, the dataset contained 11,971 records with known Quantity values, of which 1,197 cases (10%) were artificially masked. Each imputation method then attempted to reconstruct those values, and because the original values were known, it was possible to compare the estimates with the real data and objectively measure the imputation error. During this procedure, the 1,197 masked cases were added to the 604 truly missing values, resulting in a total of 1,801 records without Quantity information. This validation design made it possible to identify the best-performing imputation method before applying it to the genuinely missing cases.

15. Comparison of Imputation Methods for Quantity

To select an imputation method for the Quantity variable, three alternatives were evaluated: Predictive Mean Matching (PMM), Classification and Regression Trees (CART), and Random Forest. The comparison was conducted using artificially masked values during validation and considering accuracy, prediction error, and bias metrics.

PMM achieved an exact-match accuracy of 9.52%, which increased to 29.8% when allowing deviations of up to one unit and to 47.5% with a tolerance of two units. It also produced a mean absolute error (MAE) of 3.20, a root mean squared error (RMSE) of 3.98, and a bias of -0.016. CART achieved an exact-match

accuracy of 10.1%, slightly higher than PMM, representing a difference of 0.58 percentage points. Although PMM showed more favorable results on the remaining evaluation criteria, the available figures do not allow these differences to be quantified relative to CART and Random Forest. Overall, the results indicate limited ability to recover the exact original quantities, although the near-zero bias suggests a small average signed deviation. This does not necessarily imply greater accuracy for individual predictions; therefore, method selection should consider error metrics alongside the distributional characteristics of the imputed values.

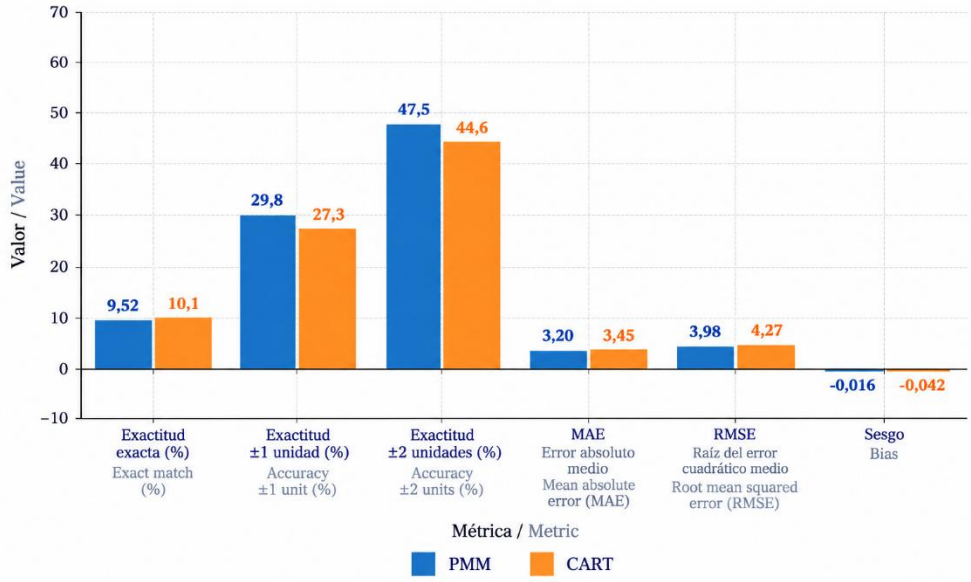


Figure 12. *Performance Comparison of Imputation Methods for the Quantity Variable.*
Source. *Prepared by the author.*

Figure 12 compares the performance of the imputation methods evaluated for the Quantity variable using accuracy, error, and bias metrics. The visualization explicitly displays the results for PMM and CART, while Random Forest was included in the overall methodological comparison but is not shown in the displayed panel. Overall, the figure shows that none of the methods can perfectly reconstruct each individual Quantity value. However, PMM provides the best overall balance between accuracy, error, and preservation of the original distribution.

Specifically, PMM achieved an exact-match accuracy of 9.52%, an accuracy of 29.8% within a margin of ±1 unit, and 47.5% within ±2 units. It also produced a mean absolute error (MAE) of 3.20, a root mean squared error (RMSE) of 3.98, and a bias of -0.016, which is very close to zero. CART, in contrast, achieved a slightly higher exact-match accuracy of 10.1%, but performed less favorably on the remaining metrics, with lower accuracy within broader tolerance ranges and higher overall prediction errors. Therefore, although CART exactly recovers a slightly larger proportion of cases, PMM provides more stable overall behavior and better preserves the structure of the data, making it the more appropriate choice for the final imputation of Quantity.

16. Preservation of the Quantity distribution

Statistic	Observed	PMM
Mean	5,66	5,64
Median	6	6
Standard deviation	2,86	2,78
Minimum	1	1
Maximum	10	10

Table 10. *Comparison of Descriptive Statistics Between the Observed and PMM-Imputed Distributions of Quantity.*
Source. *Prepared by the author.*

Table 10. Comparison of Descriptive Statistics for Quantity Between Actual Values and PMM-Imputed Values.
Table 10 compares the descriptive statistics of the Quantity variable between actual values and those obtained using Predictive Mean Matching (PMM). The mean decreased slightly from 5.66 to 5.64 units, representing a difference of only 0.02, while the median remained unchanged at 6. The standard deviation decreased from 2.86 to 2.78, indicating slightly lower variability in the imputed values. The minimum and maximum values remained at 1 and 10 units, respectively. These findings suggest that PMM approximately preserves the main characteristics of the original distribution, although a slight reduction in variability is observed. However, similarity in descriptive statistics does not guarantee preservation of the entire distribution or accuracy at the individual imputation level.

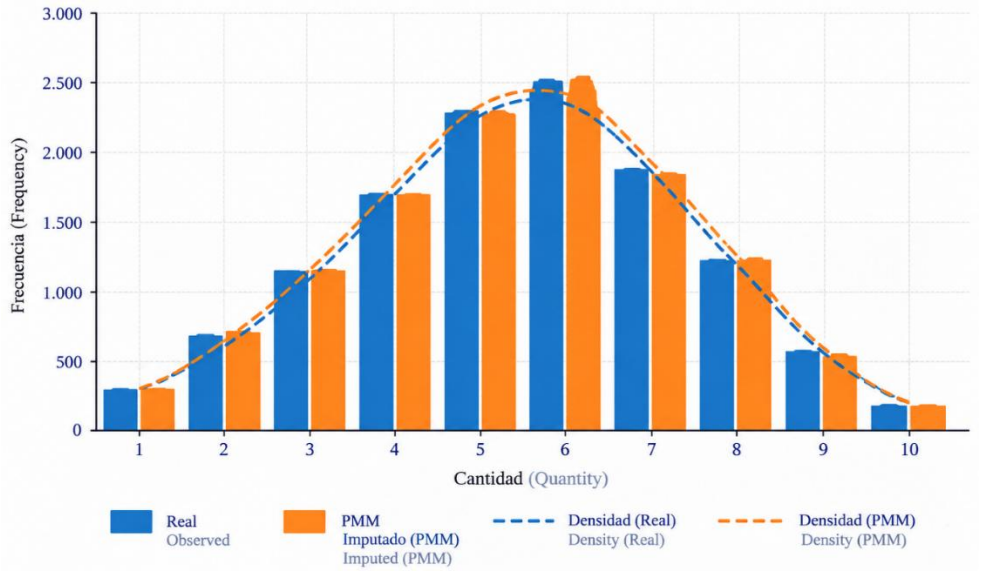


Figure 13. *Comparison of the Observed Quantity Distribution with the PMM-Imputed Distribution.*
Source. *Prepared by the author.*

Figure 13 compares the observed distribution of the Quantity variable with the distribution obtained using Predictive Mean Matching (PMM), considering the displayed frequencies and density curves. Both distributions exhibit similar shapes, with a greater concentration of observations around the central

quantities. The mean differs by 0.02 units, decreasing from 5.66 in the actual data to 5.64 in the imputed data, while the median remains unchanged at 6. The standard deviation decreases from 2.86 to 2.78, indicating a slight reduction in variability, although the minimum and maximum values remain at 1 and 10 units. These findings suggest that PMM reasonably preserves the main characteristics of the original distribution. However, similarity between the two distributions does not guarantee the accuracy of individual imputations and should therefore be interpreted alongside the error metrics obtained during validation.

17. Reconstruction of Total Spent

After completing the imputation of Quantity using Predictive Mean Matching (PMM), the missing values in Total Spent were reconstructed using the mathematical relationship between unit price and quantity purchased:

$$\text{Total Spent} = \text{Price Per Unit} \times \text{Quantity}$$

This procedure calculated all 604 missing values in Total Spent, reducing the number of missing entries from 604 to zero and achieving 100% completeness. No additional statistical method was required for this variable, as the transaction amounts were calculated directly from Price Per Unit and the imputed Quantity values. However, although the reconstruction is deterministic, the results depend on quantities previously estimated using PMM and therefore retain the uncertainty associated with that imputation. Consequently, complete recovery of Total Spent ensures consistency with the applied formula but does not necessarily guarantee the accuracy of the reconstructed amounts relative to the unknown original values.

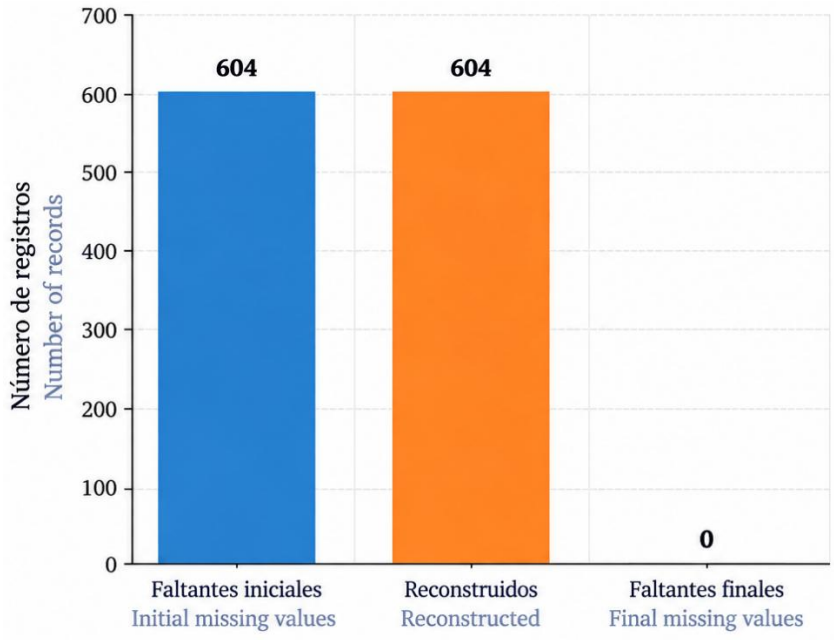


Figure 14. *Recovery of Missing Values in Total Spent.*
Source. *Prepared by the author.*

Figure 14 shows the treatment applied to the Total Spent variable after the imputation of Quantity. Initially, this variable contained 604 missing values. Once the Quantity values were obtained, it was possible to reconstruct all 604 missing Total Spent

records using the mathematical relationship between unit price and quantity purchased, so no missing values remained at the end of the process.
This result indicates that it was not necessary to build an additional statistical model for Total Spent. Instead, total spending was calculated directly by multiplying Price Per Unit by Quantity, allowing the use of a verifiable business rule and avoiding a second statistical imputation for a variable that depends on another already treated. However, it should be noted that the reconstructed Total Spent values depend on the previously imputed values of Quantity. Therefore, although the mathematical operation used for reconstruction is exact, the final result indirectly retains the uncertainty associated with the imputation of Quantity.

18. Evaluation of Discount Applied

The Discount Applied variable contained 4,199 missing values, representing 33.39% of the transactions in the original dataset. To assess whether these values could be imputed, three supervised classification models were compared: logistic regression, Classification and Regression Trees (CART), and Random Forest. Model performance was evaluated using overall accuracy and balanced accuracy, with a naive classifier achieving 50.36% accuracy as the reference.
The results showed that CART achieved the highest performance among the evaluated models, with an accuracy of 50.4% and a balanced accuracy of 50.4%. Logistic regression achieved 49.5% accuracy and 49.6% balanced accuracy, while Random Forest recorded 48.1% on both metrics. Although CART performed slightly better, its improvement over the baseline classifier was only 0.04 percentage points. This indicates that none of the models achieved a meaningful improvement in predictive performance compared with the reference.
Consequently, the results do not provide sufficient evidence to justify imputing Discount Applied using the evaluated models. Assigning values based on predictions with this level of performance could introduce classification errors and distort the original variable distribution. Therefore, retaining the missing values, documenting their presence, and avoiding imputations without sufficient predictive support is the more methodologically appropriate approach.

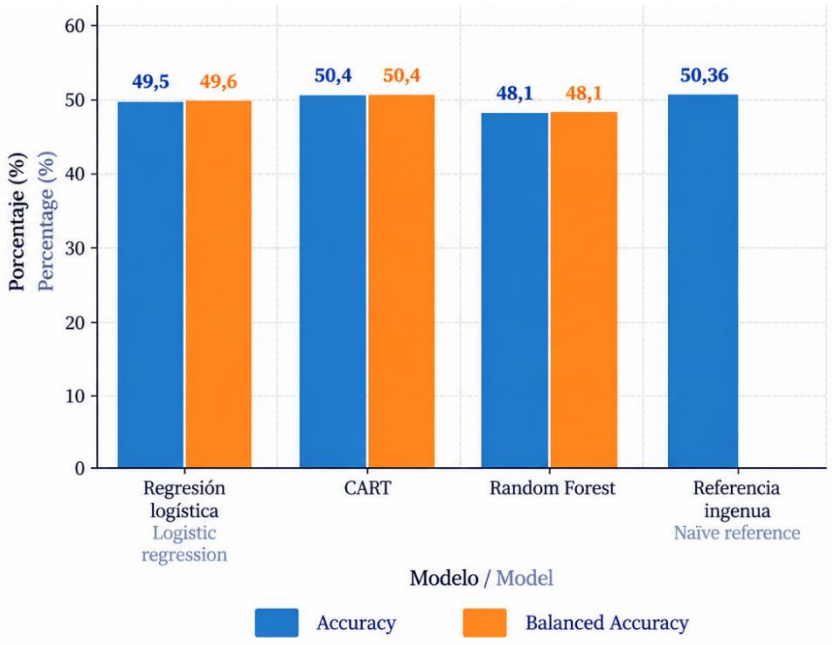


Figure 15. *Comparison of Predictive Performance of Classification Models for the Discount Applied Variable.*
Source. *Prepared by the author.*

Figure 15 presents the performance of logistic regression, CART, and Random Forest models using accuracy and balanced accuracy metrics. CART achieved the highest accuracy at 50.4%, followed by logistic regression at 49.5% and Random Forest at 48.1%. However, CART exceeded the baseline classifier, which achieved 50.36% accuracy by assigning cases to the majority class, by only 0.04 percentage points. Balanced accuracy also remained close to 50% across all three models, indicating limited ability to distinguish between transactions with and without discounts. These findings do not provide sufficient evidence to reliably impute the 4,199 missing values in Discount Applied. Consequently, these values were retained as unknown rather than assigning TRUE or FALSE labels without adequate predictive support.

19. Creation of Discount_Status

Given the limited predictive performance of the models evaluated for Discount Applied, the decision was made to retain the missing values and create a new variable named Discount_Status, with three categories: TRUE, FALSE, and UNKNOWN. This classification distinguishes transactions with discounts, transactions without discounts, and those for which insufficient information was available to determine their status. The results show that 4,219 records (33.55%) were classified as TRUE, 4,157 (33.06%) as FALSE, and 4,199 (33.39%) as UNKNOWN, accounting for all 12,575 transactions. The distribution shows similar proportions across the three categories, although UNKNOWN represents unobserved information rather than a third actual discount status. This approach preserved the uncertainty associated with the original records, avoided imputations without sufficient predictive support, and enabled missing information to be handled explicitly in subsequent analyses.

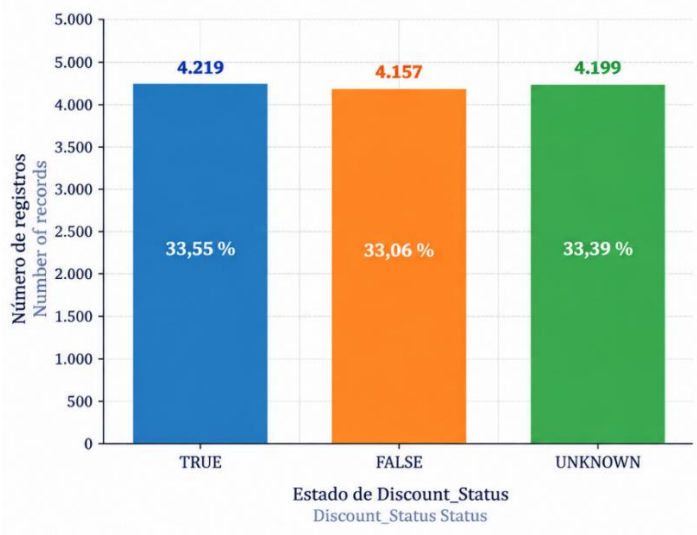


Figure 16. Distribution of the Discount_Status Variable in the Final Dataset.
Source. *Prepared by the author.*

Figure 16 shows the distribution of the Discount_Status variable, created to explicitly preserve both observed and missing information regarding discount application. The results indicate that 4,219 records (33.55%) were classified as TRUE, 4,157 (33.06%) as FALSE, and 4,199 (33.39%) as UNKNOWN. The three categories are very similar in size, each representing approximately one third of the 12,575 transactions.

It is important to note that the UNKNOWN category does not mean that no discount was applied. Its meaning is different: it indicates that there is insufficient information to

determine whether a discount was applied or not. In this way, the variable distinguishes between known observations and observations for which information is unavailable, avoiding unsupported imputations and explicitly preserving the uncertainty present in the original data.

20. Resultado general del tratamiento de los datos faltantes / Overall Results of Missing Data Treatment

Variable	Faltantes iniciales / Initial Missing Values	Tratamiento / Treatment	Resultado / Result
Item	1.213	Deterministic recovery	0 missing values
Price Per Unit	609	Deterministic reconstruction	0 missing values
Quantity	604	PMM imputation	0 missing values
Total Spent	604	Mathematical reconstruction	0 missing values
Discount Applied	4.199	No se imputa / No imputation	4,199 missing values
Discount_Status	—	TRUE/FALSE/UNKNOWN classification	0 missing values

Table 11. *Summary of Strategies Applied for Missing Data Treatment.*
Source. *Prepared by the author.*

Table 11 summarizes the results of missing data treatment through deterministic procedures, statistical imputation, and explicit preservation of unknown information. A total of 1,213 missing values in Item and 609 in Price Per Unit were recovered using business rules, while the 604 missing values in Quantity were estimated using Predictive Mean Matching (PMM). Subsequently, 604 missing values in Total Spent were reconstructed using the mathematical relationship between unit price and quantity, although these calculations retain the uncertainty associated with the imputed quantities. In contrast, the 4,199 missing values in Discount Applied were left unimputed because of the limited predictive performance of the evaluated models. To facilitate their treatment, Discount_Status was created to distinguish between applied discounts (TRUE), unapplied discounts (FALSE), and unknown information (UNKNOWN). Overall, the process achieved complete coverage for four of the five originally affected variables while explicitly preserving missing information in the remaining variable. This approach avoided assigning unsupported values and maintained a distinction between recovered data, statistically estimated values, and information that remained unknown.



Figure 17. *Methodological Workflow for Data Auditing, Cleaning, Recovery, and Imputation.*
Source. *Prepared by the author.*

Figure 17 illustrates the ten stages of the data processing workflow applied to the Retail Store Sales: Dirty for Data Cleaning dataset. The process began with a data quality audit and business rule validation, which enabled the recovery of missing values in Item and Price Per Unit through deterministic relationships. Predictive Mean Matching (PMM) was subsequently used to impute missing Quantity values,

followed by the reconstruction of Total Spent using the relationship between unit price and quantity purchased. The next stage assessed the feasibility of imputing Discount Applied, but the limited predictive performance of the evaluated models supported retaining its missing values through the UNKNOWN category in the new Discount_Status variable. Finally, the processed datasets were obtained, maintaining a distinction between observed, recovered, imputed, and unknown values. This workflow documented the transformations performed and preserved the traceability of the data treatment process.

21. Bases de datos finales / Final Datasets

Characteristic	Scientific Dataset	Operational Dataset
Records	12.575	12.575
Variables	14	13
Missing values	4.199	0
Retains Discount Applied	Sí / Yes	No
Retains Discount_Status	Sí / Yes	Yes
Recommended use	Research and reproducibility	Reporting and operational analysis

Table 12. *Comparison of the Characteristics of the Final Datasets.*
Source. *Prepared by the author.*

Table 12 presents the characteristics of the two datasets generated after completing the cleaning and imputation process. Both retain all 12,575 original transactions but differ in the number of variables and the treatment of missing information. The scientific dataset contains 14 variables and preserves the 4,199 missing values in Discount Applied, along with the Discount_Status variable, which explicitly identifies records with unknown information. This structure maintains the distinction between observed and unavailable values, making it suitable for research requiring traceability and reproducibility. In contrast, the operational dataset contains 13 variables and has no missing values because it excludes Discount Applied while retaining Discount_Status with the categories TRUE, FALSE, and UNKNOWN. This version facilitates reporting and operational analysis, provided that the UNKNOWN category is interpreted as unavailable information rather than as an indication that no discount was applied.

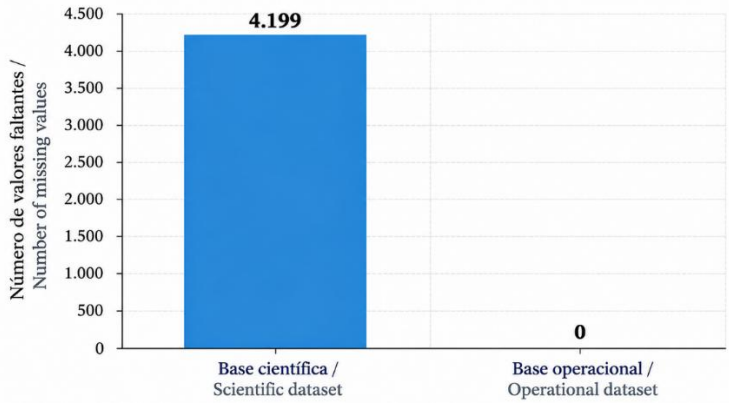


Figure 18. *Comparison of Missing Values Between the Scientific and Operational Datasets.*
Source. *Prepared by the author.*

Figure 18 shows the difference in the number of missing values between the two datasets generated after the cleaning and imputation process. The scientific dataset retains 4,199 missing values, corresponding exclusively to Discount Applied, while the operational dataset contains no missing values because this variable was excluded. Both datasets preserve all 12,575 original transactions and retain Discount_Status, which identifies records with unknown information. This difference reflects the intended purpose of each dataset: the scientific dataset explicitly preserves unobserved values to support traceability and reproducibility, whereas the operational dataset provides a structure without null values for reporting and operational analysis. However, the absence of null values in the operational dataset does not mean that all original information was recovered, since the 4,199 unknown cases remain explicitly identified through the UNKNOWN category.

22. Overall Process Summary

Process	Records Processed
Items recovered deterministically	1.213
Unit prices recovered deterministically	609
Deterministic transformations in Python	1.822
Quantity values imputed using PMM	604
Total Spent values reconstructed	604
Discount Applied values retained as unknown	4.199
Records deleted	0

Table 21. *Final Summary of the Main Transformations Performed During Data Auditing, Cleaning, and Imputation.*
Source. *Prepared by the author.*

Note: The 1,822 deterministic transformations represent the sum of 1,213 recovered Item values and 609 recovered Price Per Unit values; they are not an additional group.

Table 21 summarizes the main transformations performed during the dataset auditing, cleaning, and imputation process. In the Python stage, 1,822 deterministic recoveries were completed, including 1,213 values in Item and 609 in Price Per Unit. Subsequently, 604 missing Quantity values were imputed using Predictive Mean Matching (PMM), enabling the mathematical reconstruction of 604 missing values in Total Spent. In contrast, the 4,199 records with missing Discount Applied information were retained as unknown because of the limited predictive performance of the evaluated models. The process was completed without deleting any of the 12,575 original transactions, maintaining a distinction between recovered, statistically estimated, and unobserved values. These results indicate that the procedure improved data completeness without introducing imputations that lacked sufficient empirical support.

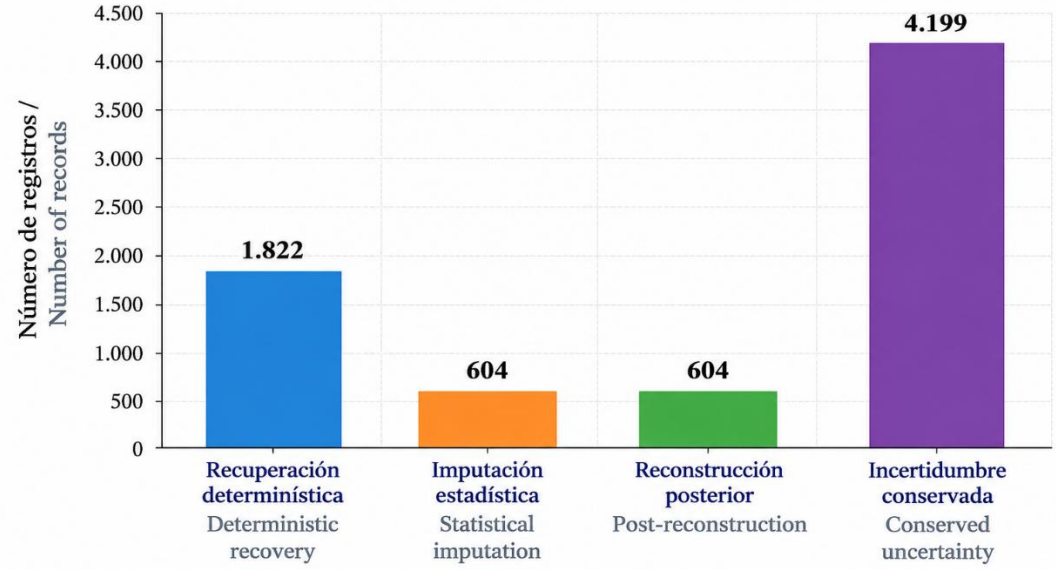


Figure 19. *Distribution of Treated Values by Procedure Applied During Data Cleaning and Imputation.*
Source. *Prepared by the author.*

Figure 19 presents the number of values treated according to the procedure applied during dataset cleaning and imputation. The largest group corresponds to Discount Applied, with 4,199 values retained as unknown because of insufficient predictive evidence to justify their imputation. During data cleaning, 1,822 deterministic recoveries were performed for the Item and Price Per Unit variables. Subsequently, 604 missing Quantity values were imputed using Predictive Mean Matching (PMM), and another 604 Total Spent values were mathematically reconstructed from the imputed quantities. These findings show that data treatment focused on distinguishing information recoverable through business rules, values requiring statistical estimation, and information that should remain unknown. It should be noted that these figures represent values treated by variable rather than necessarily distinct transactions, since a single transaction may have required more than one procedure.

23. Conclusions

The data auditing, cleaning, and imputation process applied to the Retail Store Sales: Dirty for Data Cleaning dataset improved data completeness while preserving all 12,575 original transactions, without deleting any records during processing. The methodology combined business rules implemented in Python with statistical procedures developed in R, prioritizing deterministic recovery and reserving imputation for values that could not be reconstructed exactly.

Deterministic cleaning recovered all 1,213 missing values in Item and 609 in Price Per Unit, resulting in 1,822 recoveries. These findings show that verified relationships between variables can be used to address data completeness issues without relying on statistical estimation. However, the validity of recovered values depends on the applicability of the business rules and the consistency of the information used.

For the 604 missing values in Quantity, PMM, CART, and Random Forest were compared through a validation procedure based on artificially masking known observations. PMM showed the most favorable overall performance according to the evaluated criteria, with a mean absolute error (MAE) of 3.20 units, a root mean squared error (RMSE) of 3.98, and a bias of −0.016. Additionally, the

imputed distribution exhibited characteristics similar to the observed distribution, with means of 5.64 and 5.66, respectively. Although these results supported the selection of PMM, individual imputation accuracy remained limited, indicating that estimated values should be interpreted in light of the uncertainty associated with the procedure.

After imputing Quantity, the 604 missing values in Total Spent were reconstructed using the mathematical relationship between unit price and quantity purchased. This decision avoided applying a second statistical model to a variable directly related to another that had already been estimated. Nevertheless, the reconstructed amounts retain the uncertainty associated with the imputation of Quantity, even though the calculations are mathematically consistent.

The evaluation of Discount Applied showed that the classification models had limited predictive ability, with accuracy ranging from 48.1% to 50.4%, close to the 50.36% achieved by the baseline classifier. Given the lack of sufficient predictive improvement, the decision was made not to impute the 4,199 missing values in this variable. Instead, Discount_Status was created, incorporating the UNKNOWN category to identify records whose discount status remains unknown. This decision avoided introducing artificial classifications without adequate empirical support.

As a result, two final datasets were generated: a scientific dataset containing 14 variables, which explicitly preserves missing values in Discount Applied, and an operational dataset containing 13 variables, with no null values but with unknown information identified through Discount_Status. Both retain all 12,575 original transactions and support different research and operational analysis requirements.

In conclusion, the main methodological outcome of the project was the development of a procedure that distinguishes originally observed values, values recovered through deterministic rules, statistically estimated values, and information that remains unknown. This distinction supports documentation of data provenance, recognition of imputation limitations, and prevention of the assumption that greater completeness necessarily implies greater accuracy. Therefore, the quality of the final datasets depends both on the information recovered and on the transparency with which estimates and remaining uncertainty are documented.